

Sydney Opera House

A masterpiece of 20th-century design and engineering, with its roofs rising like sails above Sydney Harbour

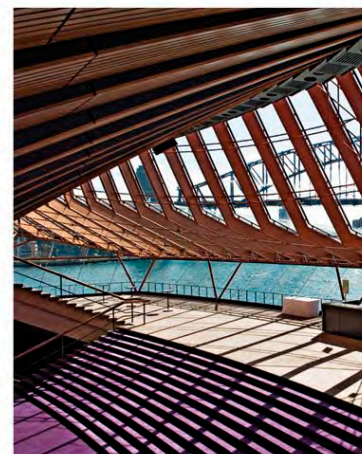
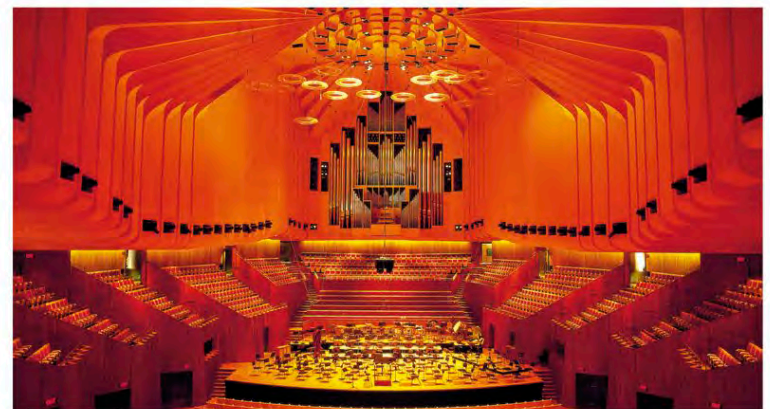


Occupying the peninsula of Bennelong Point, Sydney Opera House, with its distinctive, sail-like roofs, gives the impression of floating on the waters of Sydney Harbour. More than merely an opera house, it is a complex of performance and exhibition venues. It includes a 2,679-seat Concert Hall, the Joan Sutherland Opera Theatre, a smaller Drama Theatre, studio and recording facilities, and an outdoor forecourt performance area.

Shells of concrete

The most striking feature of the opera house is the pioneering Expressionist design, conceived by Danish architect Jørn Utzon (1918–2008). His innovative idea of a group of parabolic, precast concrete shells to form a series of curved roof arches led him to win a design competition for the site in 1957. Construction began 2 years later and was completed in 1973.

The opera house has a binary structure: two sets of interlocking parabolic arches form the roofs of the two main halls, which diverge from a single entrance foyer. These arches are made of concrete, but faced with off-white glazed tiles in a chevron pattern, to form the distinctive shells. The two main buildings sit among a series of terraced pedestrian concourses, accessed via the Monumental Steps from the shoreside forecourt.



△ PRESTIGIOUS AND MAJESTIC
No less impressive than its exterior is the fully equipped Concert Hall, the largest performance venue at the opera house. It houses the world's largest mechanical action grand organ, just above the stage.

◁ LIGHT-FILLED SPACES
Looking out over Sydney Harbour, the foyers at the front of the two sets of arches are filled with natural light from the tall, metal-framed glass frontage.

THE "SPHERICAL SOLUTION"

Utzon's original design did not specify the exact curvature of the roof shells nor how they could be constructed. Various methods of prefabricating individual concrete panels were considered, but in 1962 Utzon and his engineers had a eureka moment, realizing that all the shells could be formed from sections of a sphere with a radius of 247 ft (75m).

